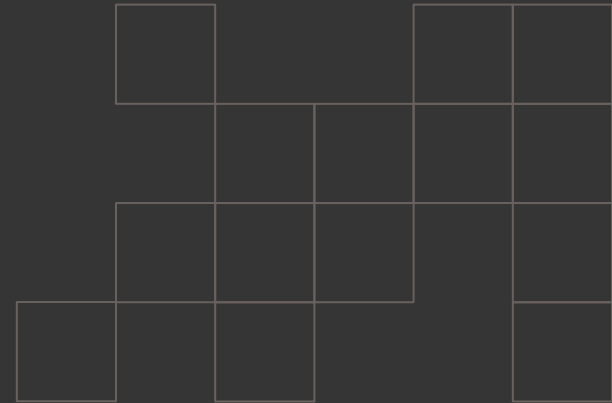
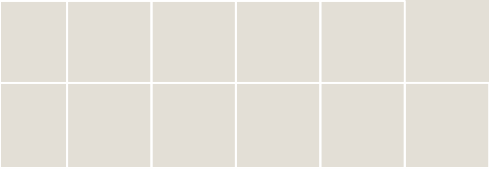


Comparing Quantum Circuit Simulators

10387 SCIQIS - August 2026





Introduction

We compare quantum circuit simulators to answer:

- How do execution time and memory scale with qubit count?
- How much does circuit structure and number of gates change the execution time?
- When does GPU acceleration become faster than CPU?



Chosen Frameworks

Framework	Vendor	Engine	GPU
Qiskit + Aer	IBM	C++ / CUDA	Yes
PennyLane Lightning	Xanadu	C++ / CUDA	Yes
Qulacs	Osaka U / QunaSys	C++ / OpenMP	No
Cirq	Google	Pure Numpy	No
Braket	Amazon	Pure Numpy	No

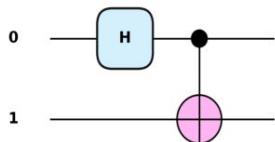
Chosen Circuits

GHZ, Random (brickwork), QFT

GHZ

H + CNOT chain

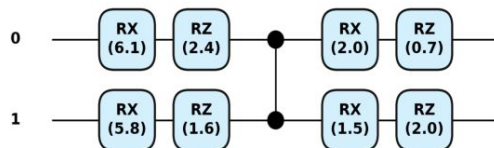
n gates
= 2 at $n=2$



Random (brickwork)

RX·RZ everywhere, then CZ
on alternating pairs, $\times n$ layers

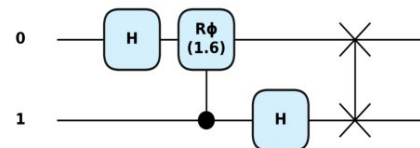
$2n^2 + n(n-1)/2$ gates
= 9 at $n=2$



QFT

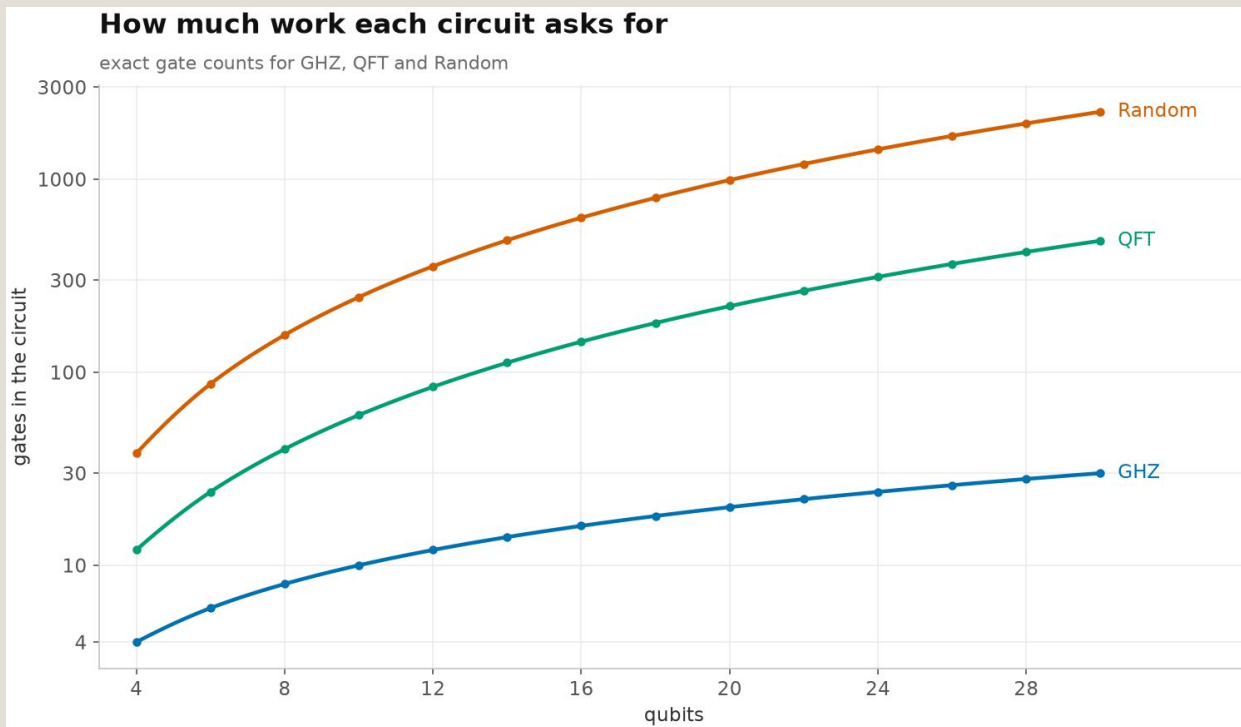
H + controlled-phase
ladder + SWAPs

$n + n(n-1)/2 + \lfloor n/2 \rfloor$ gates
= 4 at $n=2$



Chosen Circuits

GHZ, Random (brickwork), QFT



Profiling

CPU Job:

- Intel Xeon Gold 6126
- 1 core / 96GB RAM
- 4 ... 30 qubits
- All 5 frameworks

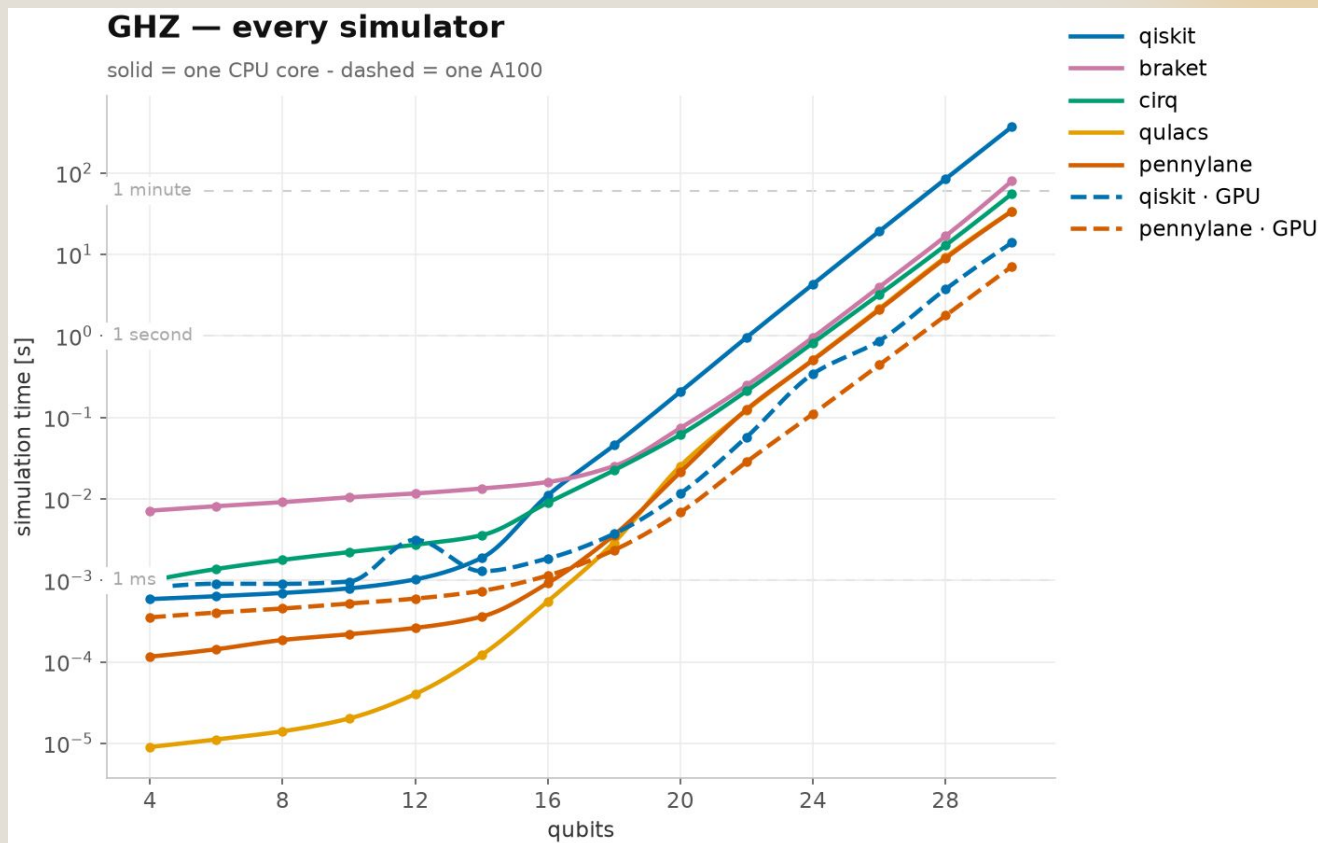
GPU Job:

- Nvidia A100
- 80GB VRAM
- 4 ... 30 qubits
- Qiskit & PennyLane

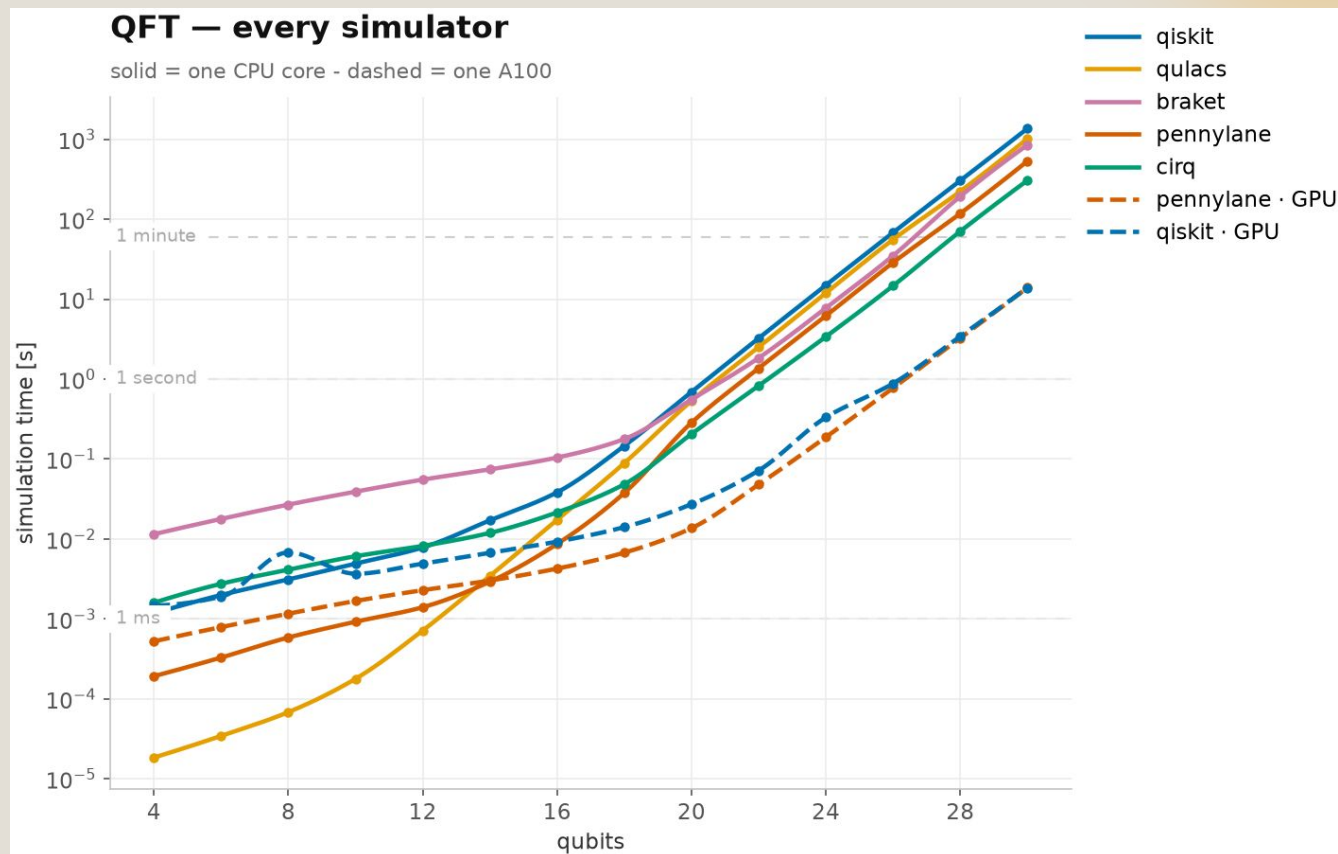
Simulation Output:

- Time of execution
- Memory used
- Gate count

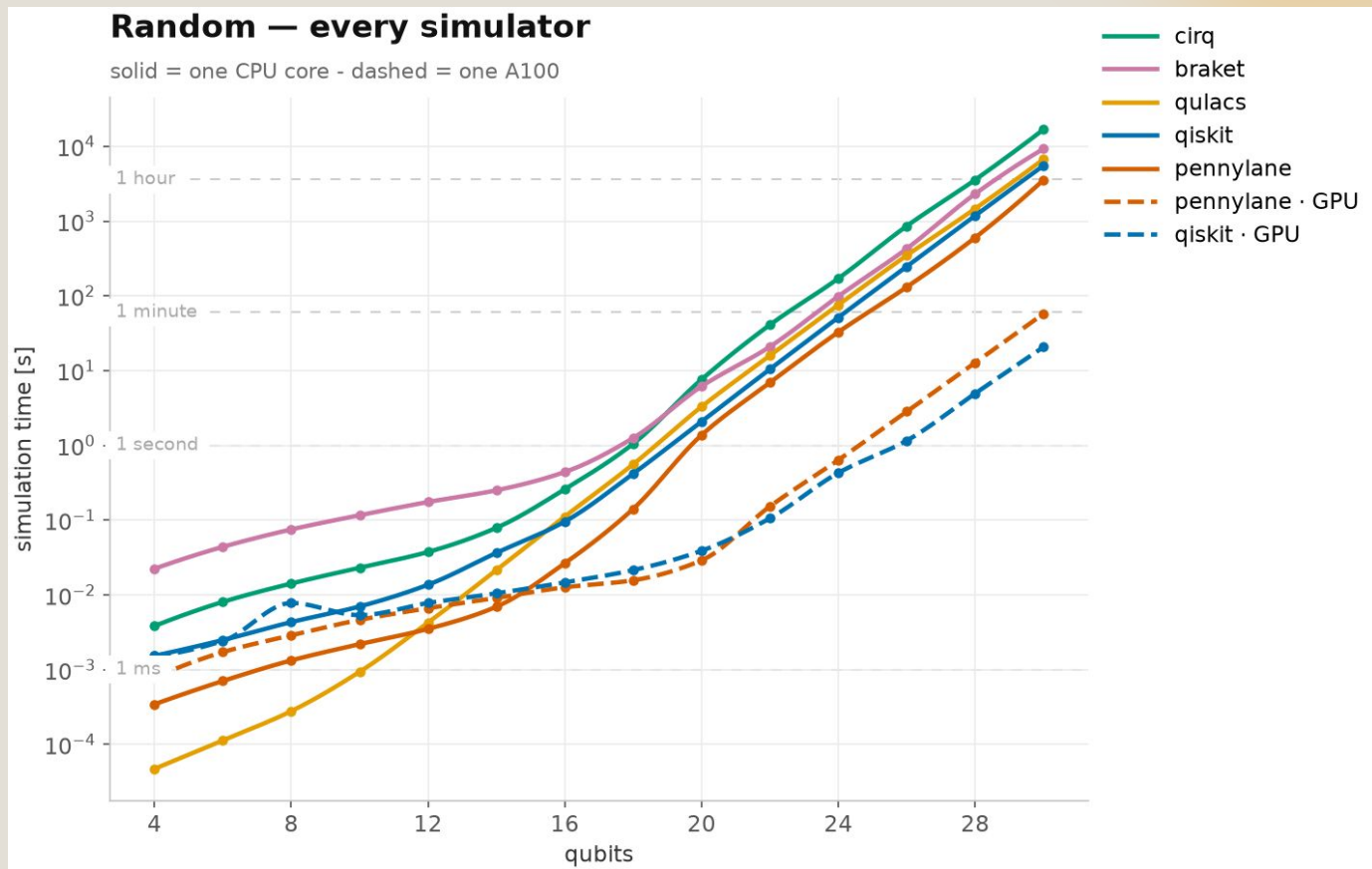
Findings: GHZ



Findings: QFT

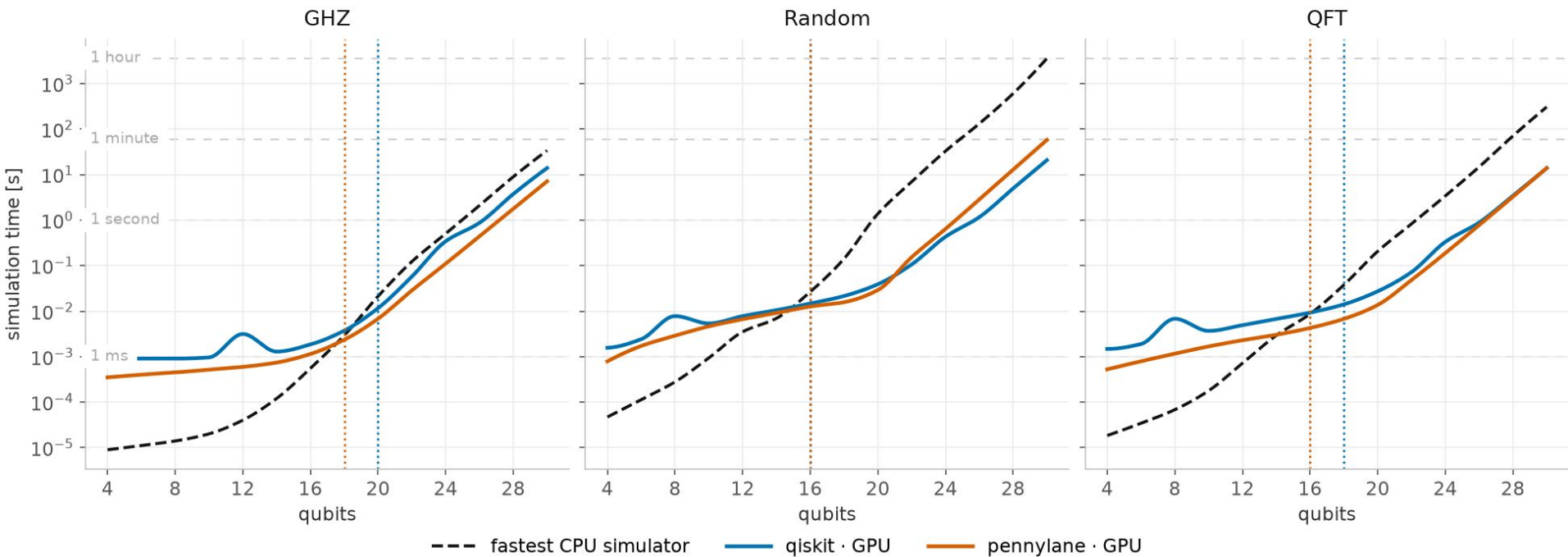


Findings: Random (brickwork)



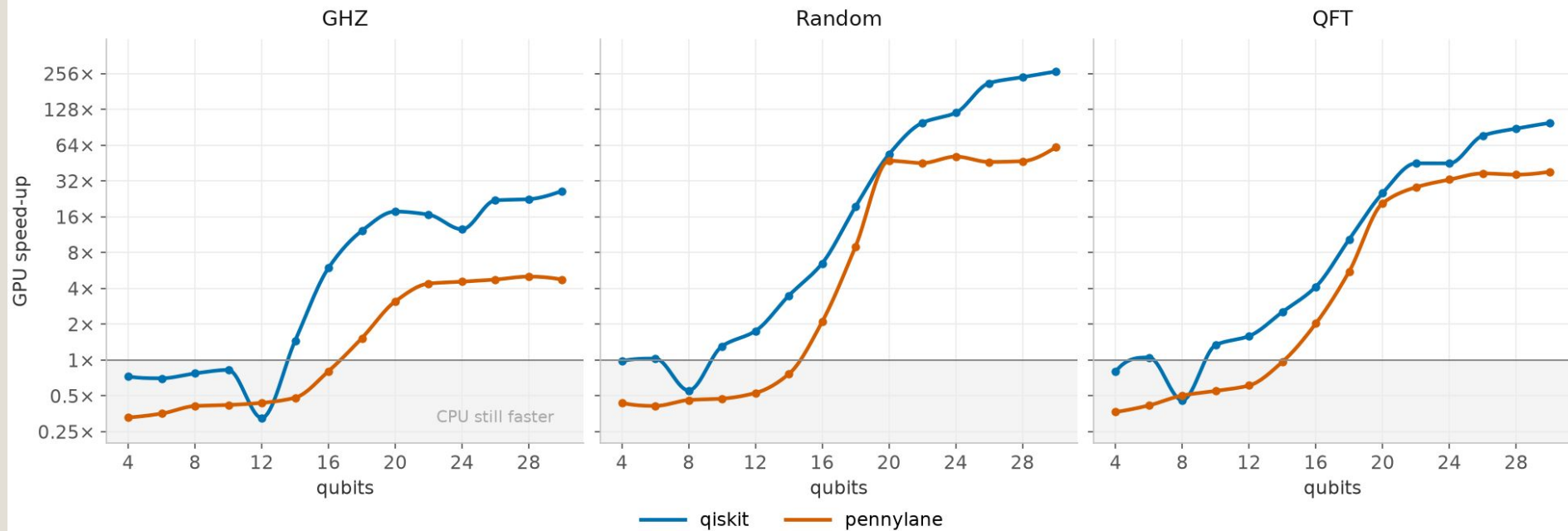
Findings: GPU crossover

Fastest CPU record vs GPU frameworks

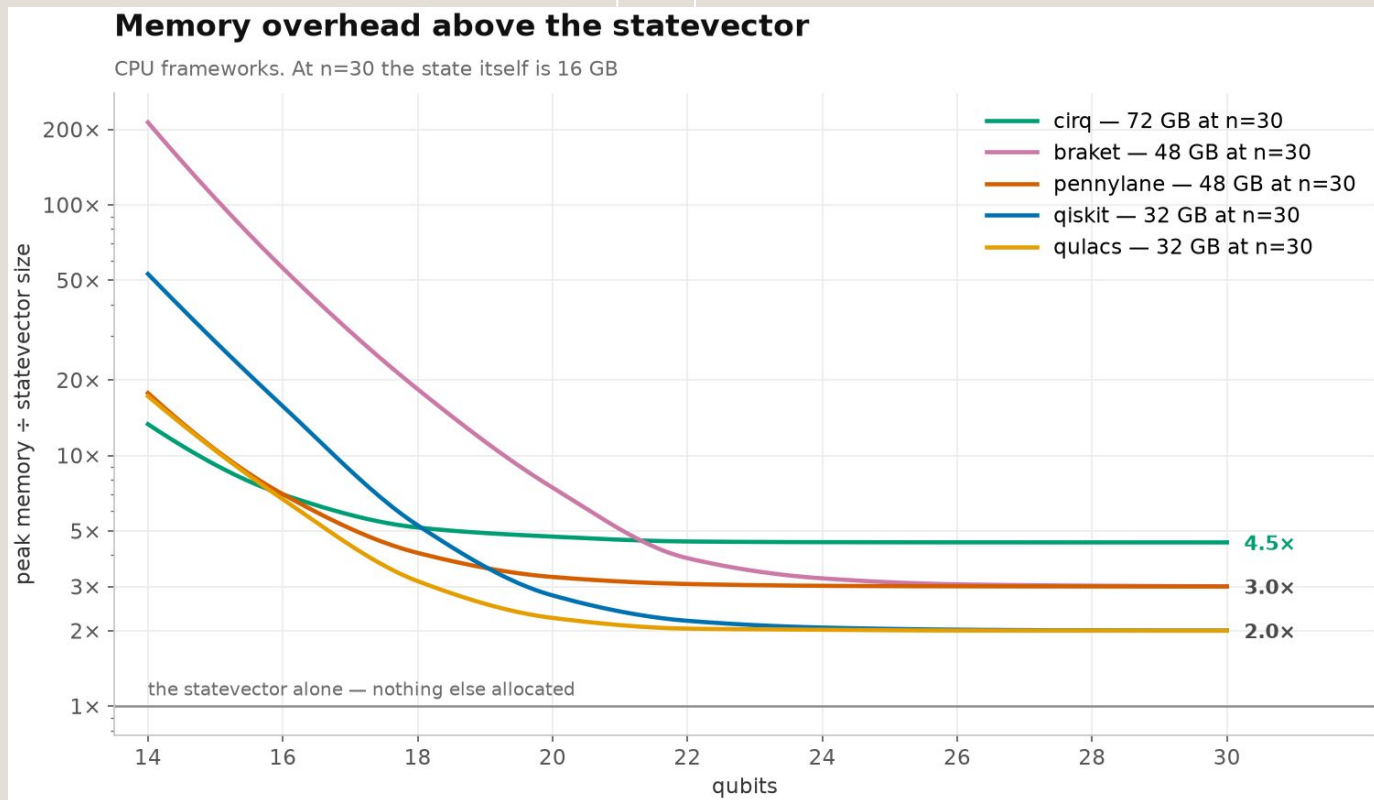


Findings: GPU speed-up

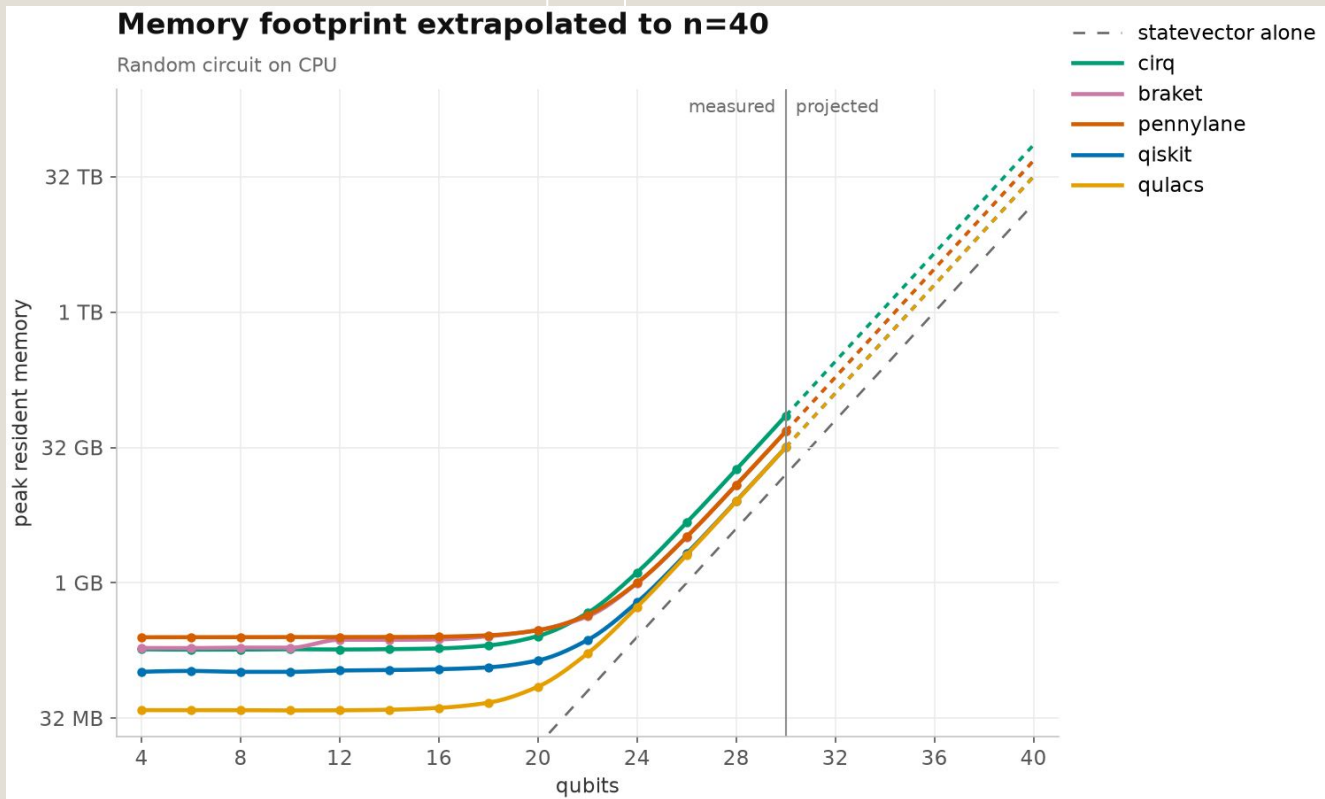
GPU speed-up over the same framework on CPU



Findings: Memory footprint



Findings: Memory footprint



Conclusions

No universal winner

The CPU ranking changes with the circuit: PennyLane leads Random, Cirq leads QFT, and PennyLane and Qulacs are strongest on GHZ.

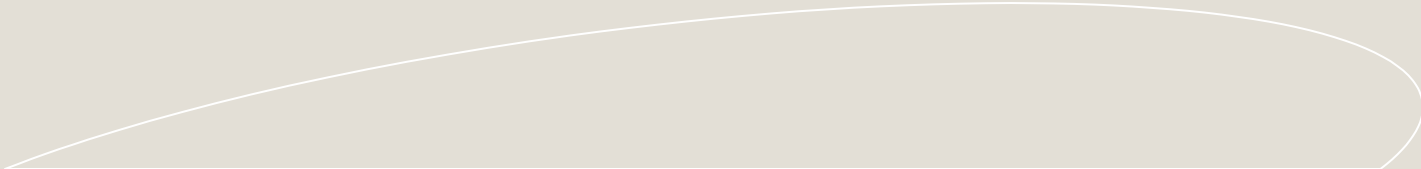
GPU acceleration has a crossover

Against the best CPU result, the A100 becomes worthwhile at approximately 16–20 qubits; beyond that, the speedup becomes substantial.

The statevector is only a memory lower bound

For Random at (n=30), peak CPU memory ranges from approximately 32 to 72 GiB

Choose by **circuit (gates count)**, **qubit count**, hardware, and **memory budget**. not by framework.





Thank you